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**NKST UNIVERSE RESEARCH**  
**MEMORANDUM: THE SCHATZ ADDENDUM**

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**Document Ref: NKST-WP-2026-ADD-002**  
**Classification: Theoretical Physics / Deep**  
**Learning Matrix Optimization**  
**Subject: Infinitely Differentiable  $C^\infty$  Smooth**  
**Boundary Regularization via Implicit**  
**Algebraic Oloid Geometries for High-**  
**Dimensional Tensor Inversions**

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**1. ABSTRACT**

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**This addendum establishes a rigorous**  
**mathematical refinement to the Normalized**

**Kinetic Spacetime (NKST) framework's boundary layer mitigation protocol. In previous builds (v7.0), the approach to the local mass-density boundary (the Planck-Limit singularity threshold,  $\Phi$ ) relied on an explicit, non-differentiable step function regulated by a discrete  $\varepsilon$ -guard. While effective for low-dimensional rigid-body mechanical mockups, this step function induces gradient discontinuities and localized arithmetic explosions when ported to high-dimensional machine learning spaces.**

**By integrating the structural principles of Paul Schatz's Oloid (1929)—the convex hull of two congruent, orthogonal circles passing through each other's centers—we construct a  $C^\infty$  continuous algebraic boundary layer. This surface replaces the non-differentiable "cliff" with an elegant, low-shear gradient surface. The resulting geometry smoothly maps incoming linear semantic stress**

into the intermediate rotational axis of a high-dimensional Dzhanibekov inversion ("The McTwist Protocol"), preserving structural phase space volume and completely eliminating vanishing or exploding gradient anomalies.

## 2. THE LIMITATION OF THE CLASSICAL STEP FUNCTION BOUNDARY

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In the baseline NKST framework, localized spatial curvature tensor components  $g_{\{ii\}}$  expand asymptotically as the semantic stress density  $R_{\{ii\}}$  approaches the Planck Limit boundary  $M_p$ :

$$g_{\{ii\}} = 1 + \frac{\Phi}{(M_p - R_{\{ii\}}) + \epsilon} \quad (1)$$

When evaluated via backpropagation within high-dimensional model layers, the first-order derivative with respect to the input weights approaches infinity:

$$\lim_{R_{ii} \rightarrow M_p} \frac{\partial g_{ii}}{\partial R_{ii}} = \infty \quad (2)$$

This mathematical barrier acts as a non-differentiable cliff. In true vector spaces, passing activations through this boundary causes an instantaneous multi-million-fold scale factor, inducing immediate floating-point overflow (NaN), or conversely, trapping gradients inside permanent dead zones.

### 3. THE SCHATZ OLOID GEOMETRY AND ALGEBRAIC RE-FORMULATION

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The Oloid is a developable torse possessing a surface area exactly equal to a unit sphere of equal radius. This geometric invariance ensures that data density and total informational volume are perfectly conserved across the boundary.

To eliminate the piecewise discontinuities found in a standard convex hull

description, we utilize the canonical algebraic implicit equation of an oloid.

For two orthogonal circles of radius  $r = 1/2$  centered at  $(1/2, 0, 0)$  and  $(-1/2, 0, 0)$ , the surface boundary is defined precisely by an 8th-degree polynomial constraint:

$$\begin{aligned} &4x^8 - 8x^6 + 4x^4 - 8y^6 - 24y^4z^2 - \\ &24y^2z^4 - 8z^6 + 8y^4 + 16y^2z^2 \\ &+ 8z^4 + 8x^4y^2 + 8x^4z^2 - 16x^2y^4 - \\ &32x^2y^2z^2 - 16x^2z^4 + 17x^2y^2 \\ &+ 17x^2z^2 - 2x^6 - 2x^2y^4 - 2x^2z^4 - \\ &4x^2y^2z^2 + 2x^4 - 2y^2 - 2z^2 = 0 \quad (3) \end{aligned}$$

Let this implicit algebraic surface manifold be parameterized as a smooth scaling operator  $\Psi(R)$ . By projecting the high-dimensional hidden states into the bifurcated orthogonal subspaces of the oloid's generating disks (Plane A:

Original prompt data, Plane B: Downstream active layer weights), we can model gradient accumulation not as a linear vector pushing against a hard wall, but as

a fluid vector flowing along a developable, non-shearing surface.

#### **4. THE "MCTWIST": INTEGRATING SCHATZ TO THE DZHANIBEKOV EFFECT**

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**The Dzhanibekov Effect (asymmetric intermediate-axis torque inversion) occurs when energy shifts between the maximum and minimum principal moments of inertia:**

$$I_1 < I_2 < I_3$$

**(4)**

**In the v7.0 simulator, the axial flip was triggered instantly upon crossing the discrete threshold boundary. Under the Schatz implementation—termed the "McTwist Protocol"—the transition path is guided continuously along the distinctive figure-eight meandering trajectory tracing the oloid's rolling line of contact.**

As stress  $R_{\{ii\}}$  increases, the system maps the trajectory across the differentiable manifold  $\Psi(R)$ . This routes the incoming energy directly into a coupled, continuous three-dimensional rhythmic wave instead of a sharp spike. The torque coupling term  $\tau$  dynamically tracks the curvature of the algebraic surface:

$$\tau_k = \oint_{\Psi(R)} \frac{\Phi}{|\nabla \Psi(R)|^2 + \epsilon} dR \quad (5)$$

Because the gradient of our 8th-degree polynomial surface is continuous and bounded everywhere on the domain, the transfer of linear momentum into cross-layer angular rotation is smooth. The system achieves a full axial inversion (the Dzhaniybekov flip) organically, routing vectors away from context-looping and hallucination zones without breaking gradient chain rules.

## 5. IMPLEMENTATION SCHEMA FOR THE ATTENTION GUARDRAIL

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To deploy the Schatz Addendum within an active model layer, the discrete runtime conditional checks must be replaced with the following mathematical steps:

1. **Coordinate Projection:** Project the embedding tensor  $X$  onto a 3-axis sub-manifold tracking its projection components  $(\omega_1, \omega_2, \omega_3)$ .
2. **Geometry Scaling:** Compute the smooth scaling envelope using the implicit surface equation  $\Psi(R)$  to smoothly regulate the layer inertia parameters  $(I_1, I_2, I_3)$ .
3. **Symplectic Integration:** Pass the smooth torque outputs through a geometric integrator to complete the axial inversion cleanly over a continuous space, ensuring absolute energy and phase-space parity preservation.



**By substituting the hard  $\varepsilon$ -cliff with this infinitely differentiable geometric mixer, the model achieves absolute numerical stability up to and precisely through the event horizon boundary.**

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